

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 5: Arithmetic Progressions

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Sum of the first n terms

1. An A.P. has first term $a = 7$ and n th term $a_n = 84$, and its first n terms add up to $\frac{2093}{2}$. Find n . (a) 22 (b) 24 (c) 23 (d) 26 **CBSE 2024, 1 mark**

SOLUTION

$$S_n = \frac{n}{2}(a + a_n) = \frac{n}{2}(7 + 84) = \frac{91n}{2}$$

$$\frac{91n}{2} = \frac{2093}{2} \text{ gives } n = 23$$

Answer: (c)

2. In an A.P., the first and eighth terms have sum 32 and product 60. Find the first term and the common difference, and then the sum of the first 20 terms. **CBSE 2024, 5 marks**

SOLUTION

$$a + (a + 7d) = 32, \text{ so } a + 7d = 32 - a.$$

$$\text{Product: } a(32 - a) = 60, \text{ so } a^2 - 32a + 60 = 0 \text{ and } (a - 2)(a - 30) = 0. \text{ Hence } a = 2 \text{ or } a = 30.$$

$$\text{If } a = 2: 7d = 28, \text{ so } d = 4. \text{ Then } S_{20} = \frac{20}{2}[2(2) + 19(4)] = 10 \times 80 = 800.$$

$$\text{If } a = 30: 7d = -28, \text{ so } d = -4. \text{ Then } S_{20} = \frac{20}{2}[2(30) + 19(-4)] = 10 \times (-16) = -160.$$

Answer: $a = 2, d = 4, S_{20} = 800$; or $a = 30, d = -4, S_{20} = -160$.

3. An A.P. has 40 terms. Its first 9 terms add up to 153 and its last 6 terms add up to 687. Find the first term, the common difference and the sum of all 40 terms. **CBSE 2024, 5 marks**

SOLUTION

$$S_9 = \frac{9}{2}[2a + 8d] = 153, \text{ so } a + 4d = 17.$$

$$\text{The last 6 terms are } a_{35} \text{ to } a_{40}. \text{ Their sum} = \frac{6}{2}(a_{35} + a_{40}) = 3(2a + 73d) = 687, \text{ so } 2a + 73d = 229.$$

$$\text{Substituting } a = 17 - 4d: 34 - 8d + 73d = 229, \text{ so } 65d = 195 \text{ and } d = 3. \text{ Then } a = 5.$$

$$S_{40} = \frac{40}{2}[2(5) + 39(3)] = 20 \times 127 = 2540$$

Answer: $a = 5, d = 3$, and the sum of all 40 terms is 2540.

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